Car Price Prediction

Milestone 2: Data Preparation (v2 — 3 Fixes Applied)

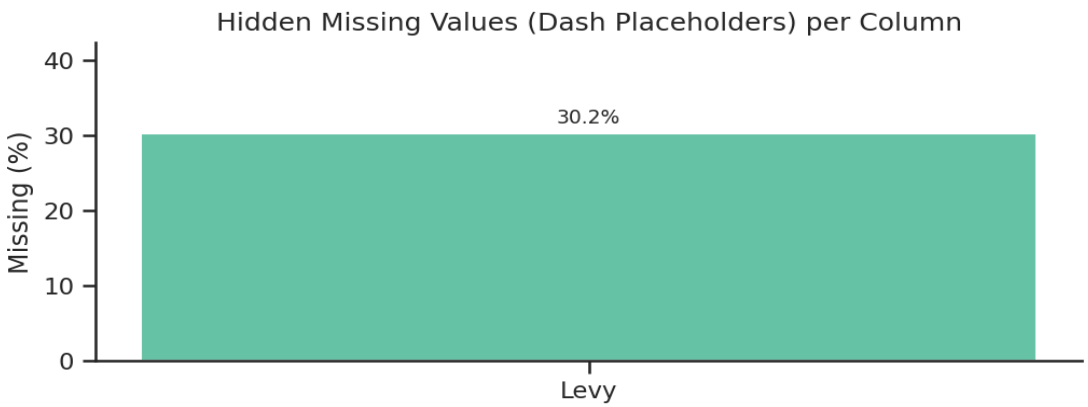
INTRODUCTION AND REPRODUCIBILITY

The objective of this milestone was to transform the raw car listings dataset into a fully clean, engineered, and machine-learning-ready form. A fixed random seed (**RANDOM_SEED = 42**) is used throughout for full reproducibility. This version corrects three issues identified in the initial submission.

CORRECTIONS FROM INITIAL SUBMISSION

#	Issue	Fix Applied
1	Ordinal encoding on Manufacturer/Model creates fake numeric ordering	Replaced with **Target Encoding** (mean Price per category)
2	Threshold importance > 0 kept all 36 features — no real selection	Applied threshold **importance > 0.002** for genuine dimensionality reduction
3	Manual preprocessing steps — error-prone and hard to reuse	Wrapped into **sklearn Pipeline + ColumnTransformer**

IMPROVING DATA QUALITY

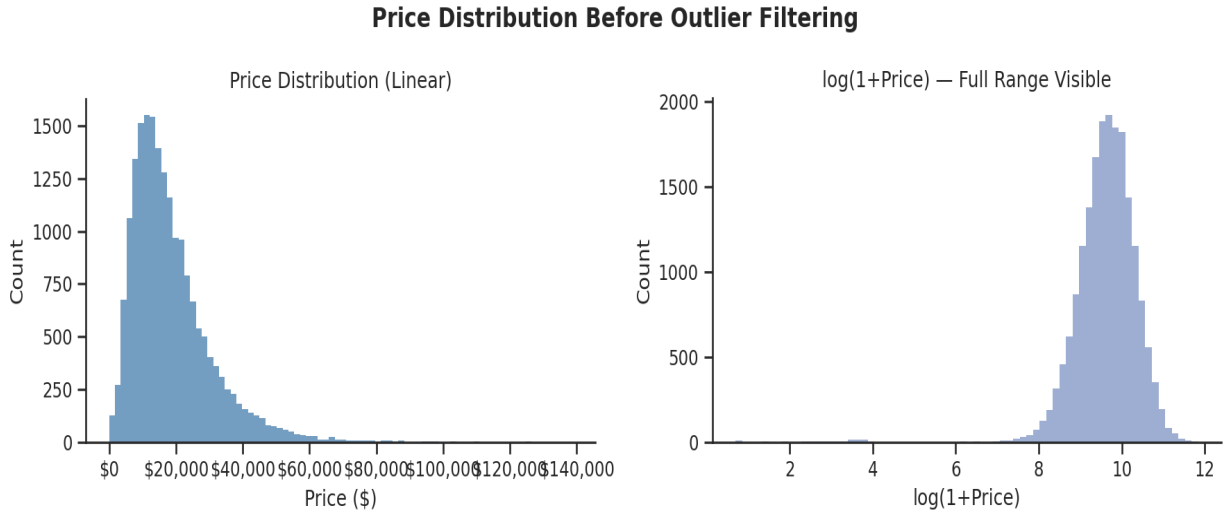


Hidden dash-placeholder missing values per column. Levy is most affected (~30%).

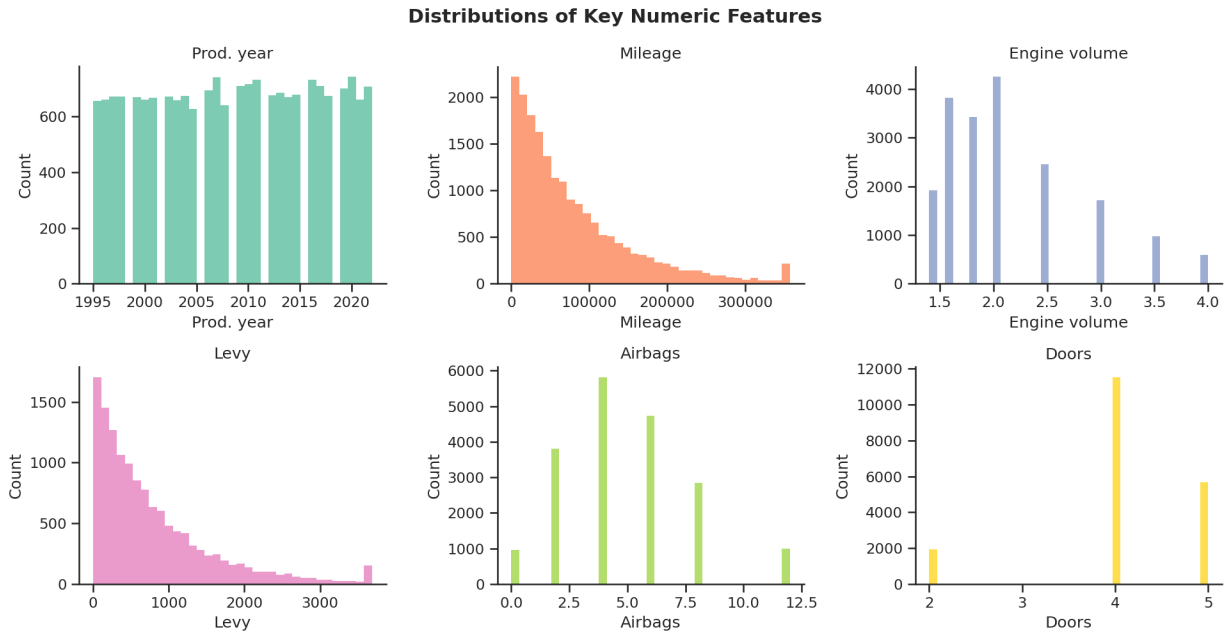
- **Dash replacement:** All "-" entries standardised to NaN before type conversion.
- **Mileage:** "186005 km" → 186005 by stripping the unit suffix.
- **Engine volume:** "2.0 Turbo" → 2.0 via regex; "Turbo" stored as a separate binary feature.
- **Levy:** Imputed with **median** — chosen over mean because Levy is right-skewed.

- **Price:** Rows where Price could not be parsed were dropped (target cannot be imputed).

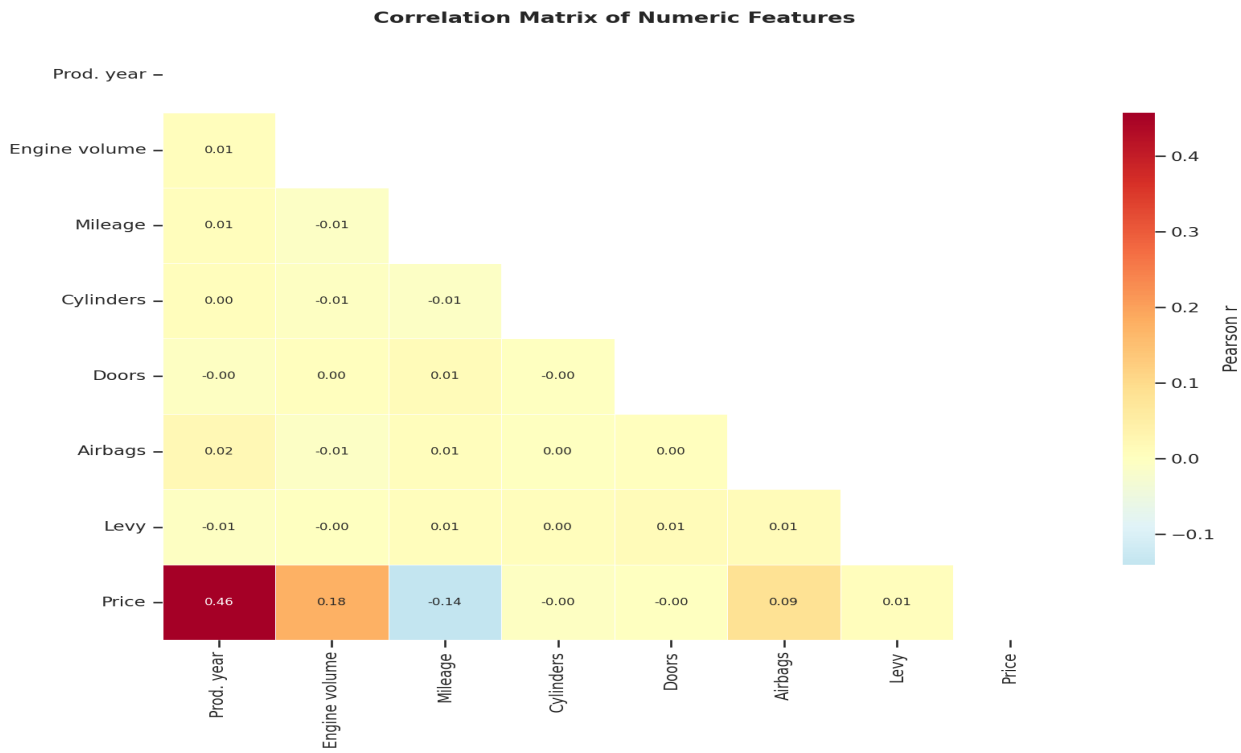
DISTRIBUTIONS



Price distribution — linear scale (left) and log scale (right).

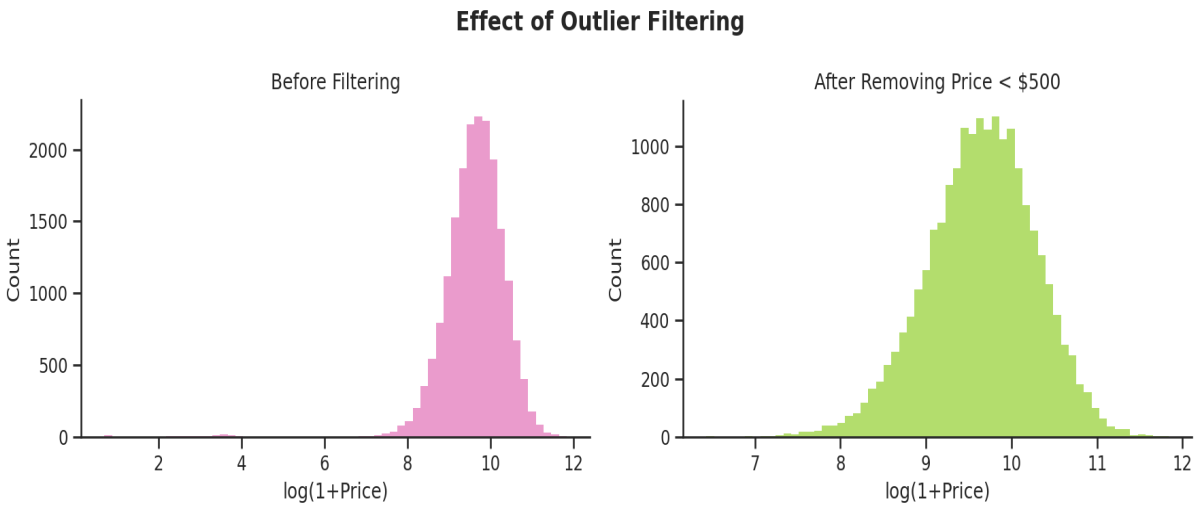


Distributions of six key numeric features across the cleaned dataset.



Full correlation matrix of all numeric features. Production Year and Mileage are the dominant predictors.

OUTLIER INVESTIGATION AND FILTERING

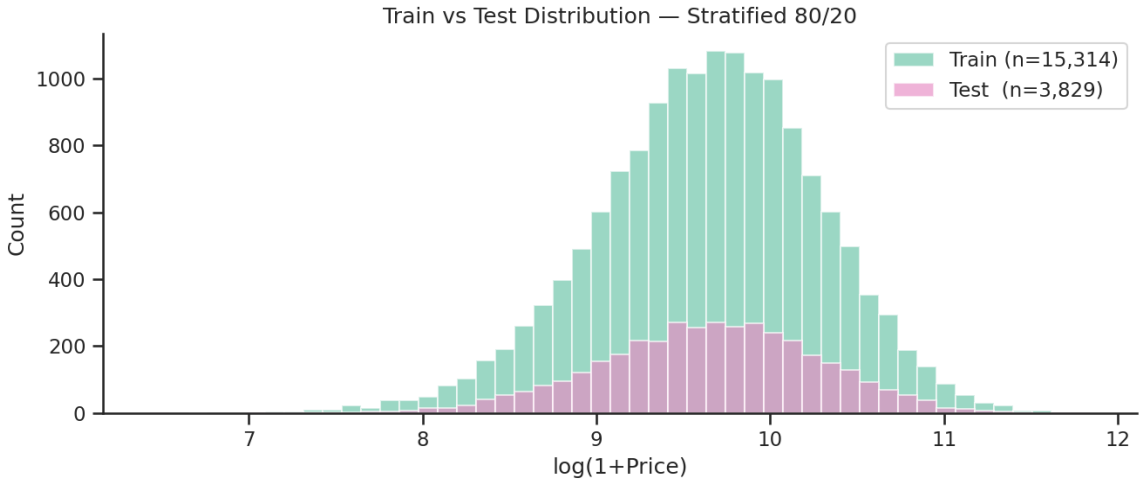


Price distribution before and after filtering. Right tail (luxury cars) is preserved.

- **Lower bound ($500):** Prices like $1 or $2 are clear data entry errors with no real-world meaning.
- **No upper bound:** Luxury vehicles are valid data points. Removing them would create systematic underestimation bias.

PREPARATION AND DATA SPLITTING

Order of operations: The split is performed before any feature engineering or scaling. The dataset is split 80% training / 20% test, **stratified by price decile** to ensure both sets reflect the same price composition.

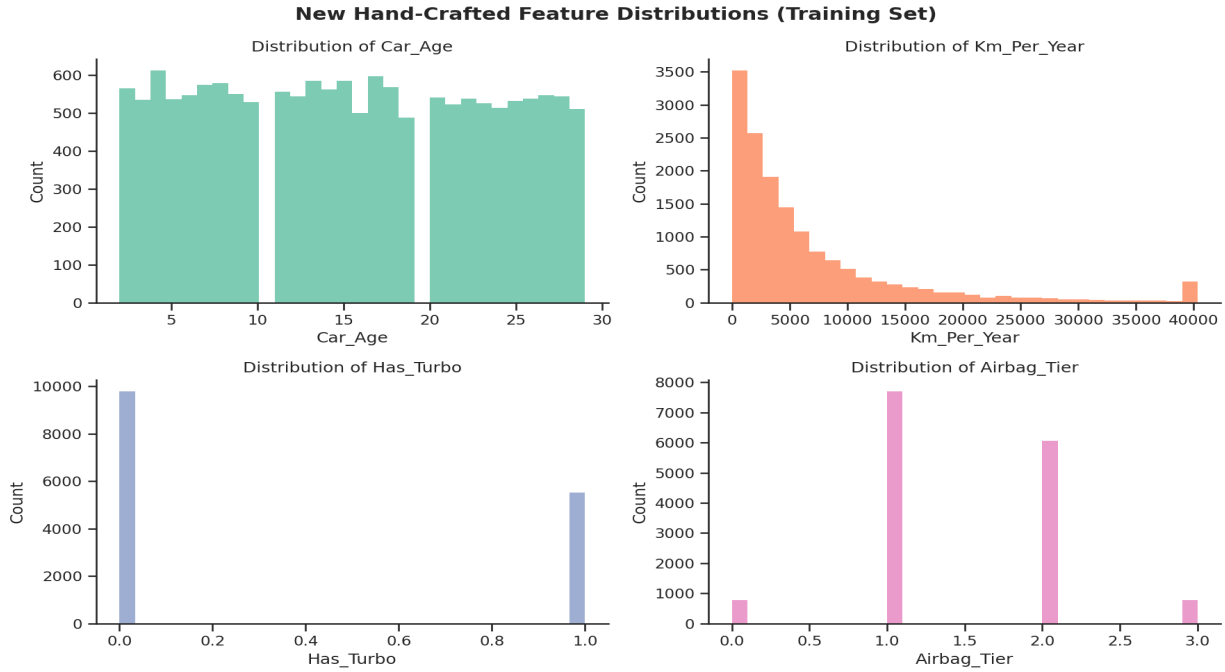


Train vs Test log-price distributions — near-identical shapes confirm the stratification worked.

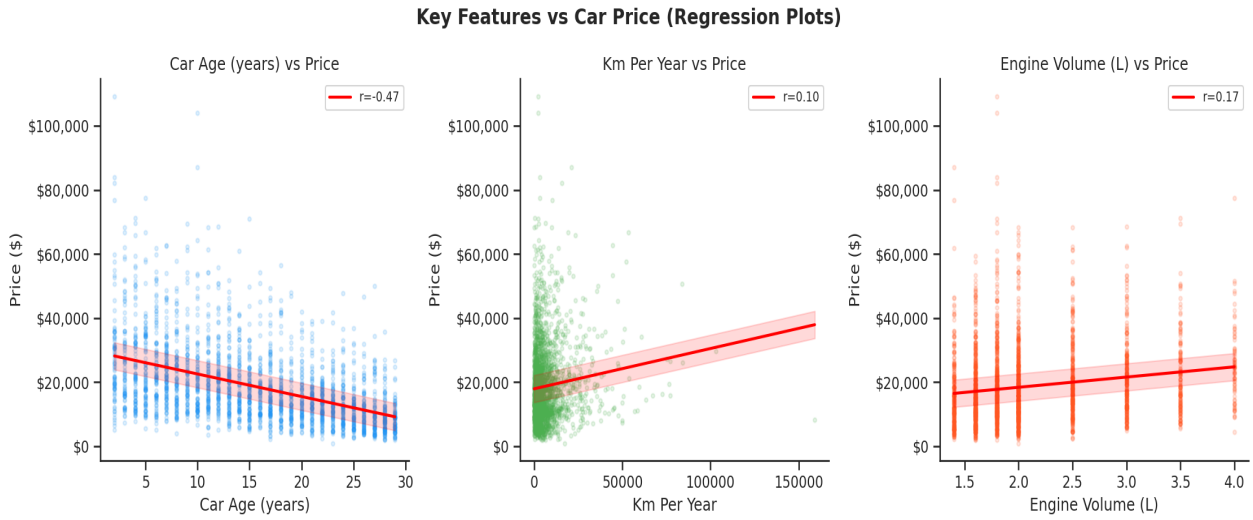
FEATURE ENGINEERING

Feature	Formula	Reasoning
Car_Age	2024 – Prod. year	Captures depreciation timeline directly
Km_Per_Year	Mileage ÷ (Car_Age + 1)	Normalises wear by ownership duration
Has_Turbo	1 if "Turbo" in raw string	Turbo engines command a price premium
Airbag_Tier	Binned: 0→T0, 1-4→T1, 5-8→T2, 9+→T3	Safety tier reduces outlier airbag effect

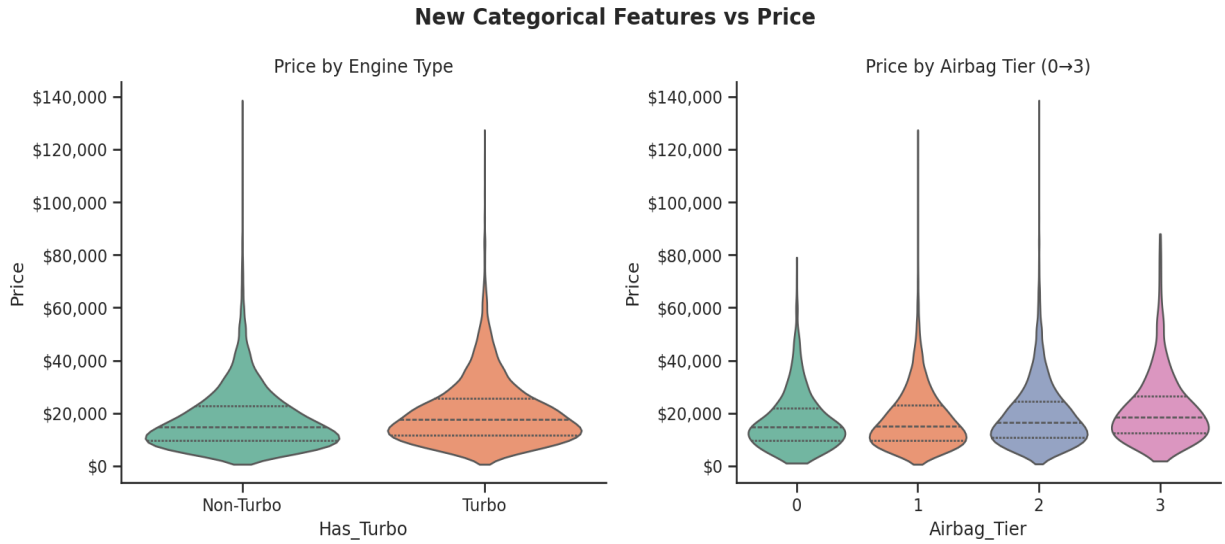
ANALYSIS OF NEW FEATURES



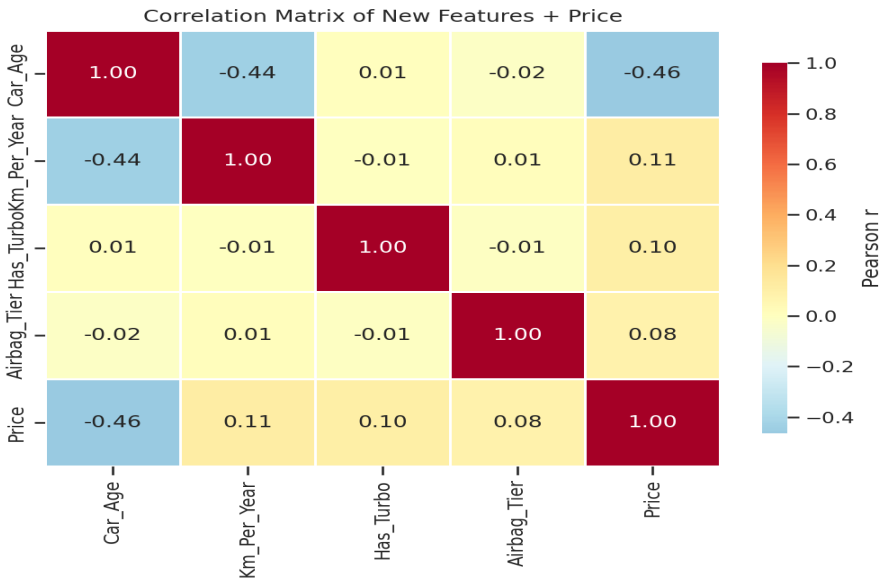
Distributions of all four new features. Car_Age is right-skewed; Km_Per_Year is heavily right-skewed.



Regression plots with trend lines. Car Age has a clear negative slope (r ≈ -0.5); Engine Volume is positive.



Violin plots — turbo engines and higher airbag tiers both correspond to higher median prices.



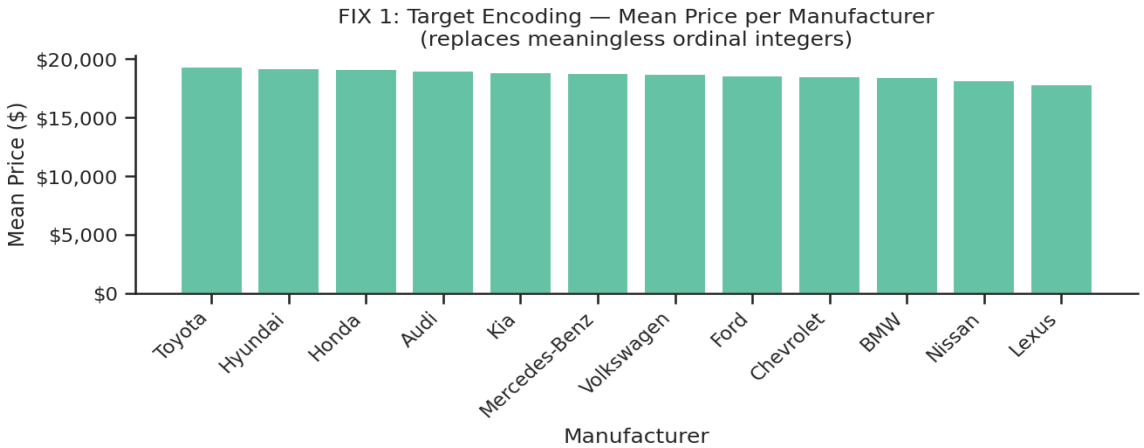
Correlation matrix of new features + Price. Car_Age and Km_Per_Year show the expected negative correlations.

FIX 1: TARGET ENCODING FOR HIGH-CARDINALITY FEATURES

■ Fix 1 — Encoding

Problem: Ordinal encoding (Toyota=0, BMW=1, Ford=2) creates a fake numeric order. The model incorrectly learns Toyota < BMW < Ford.

Fix applied: Target Encoding replaces each category with the mean Price of that category in the training set. BMW → ~$45,000 because that is what BMWs actually sell for. Meaningful signal, not fake ranking.



Target encoding result — each manufacturer mapped to its mean sale price in the training set.

One-Hot Encoding was retained for low-cardinality features (Fuel type, Gear box type, etc.) since these have few unique values and OHE avoids any ordering assumptions.

FIX 3: sklearn PIPELINE + COLUMNTRANSFORMER

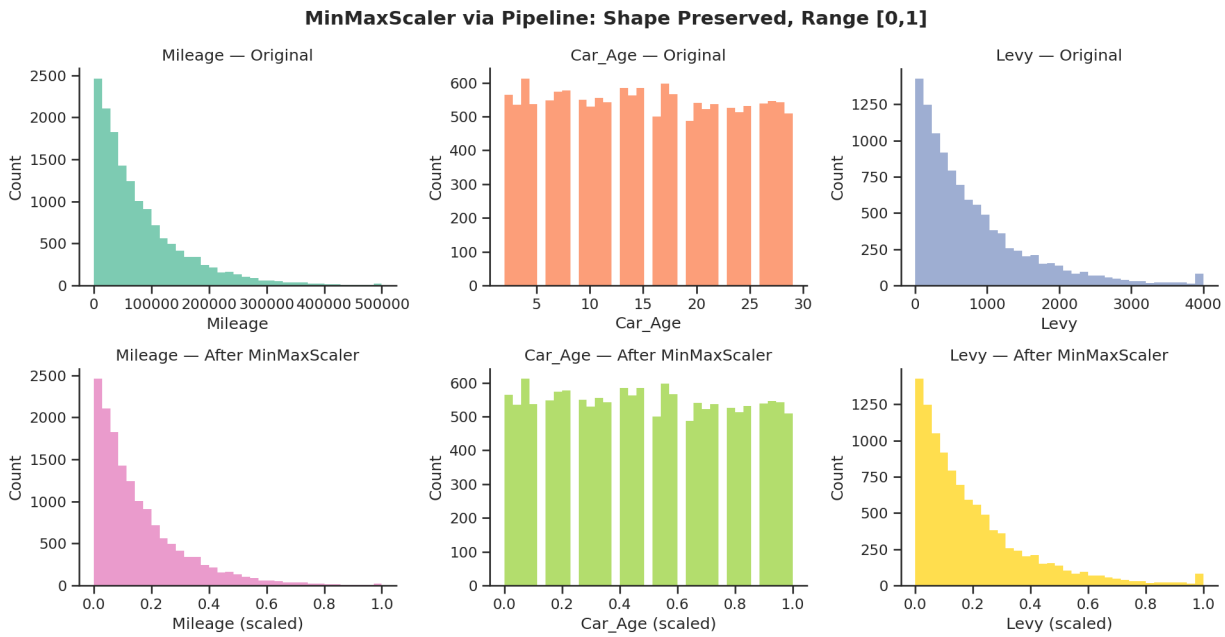
■ Fix 3 — Pipeline

Problem: Manual preprocessing steps are error-prone: easy to accidentally fit the scaler on both train and test, or apply steps in the wrong order.

Fix applied: ColumnTransformer + Pipeline wraps all scaling steps. The pipeline is fit once on training data only, then transform() is called on the test set — leakage is structurally impossible.

The pipeline structure is:

- **Step 1 — ColumnTransformer:** Applies MinMaxScaler to all numeric columns.
- **Step 2 (M3):** A regression model will be appended as the final pipeline step for seamless training and prediction.



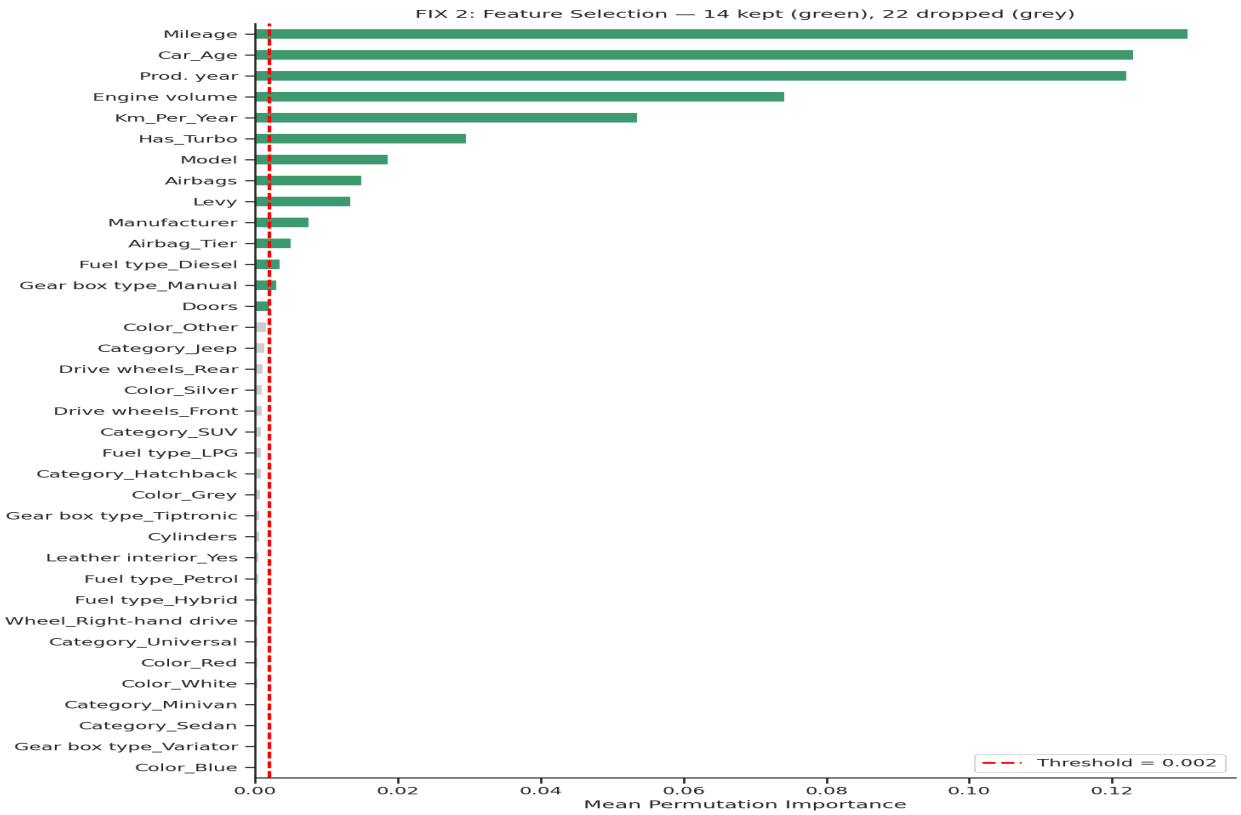
MinMaxScaler applied via Pipeline. Shape is preserved; range normalised to [0,1]. Scaler fitted on training only.

FIX 2: FEATURE SELECTION WITH REAL THRESHOLD

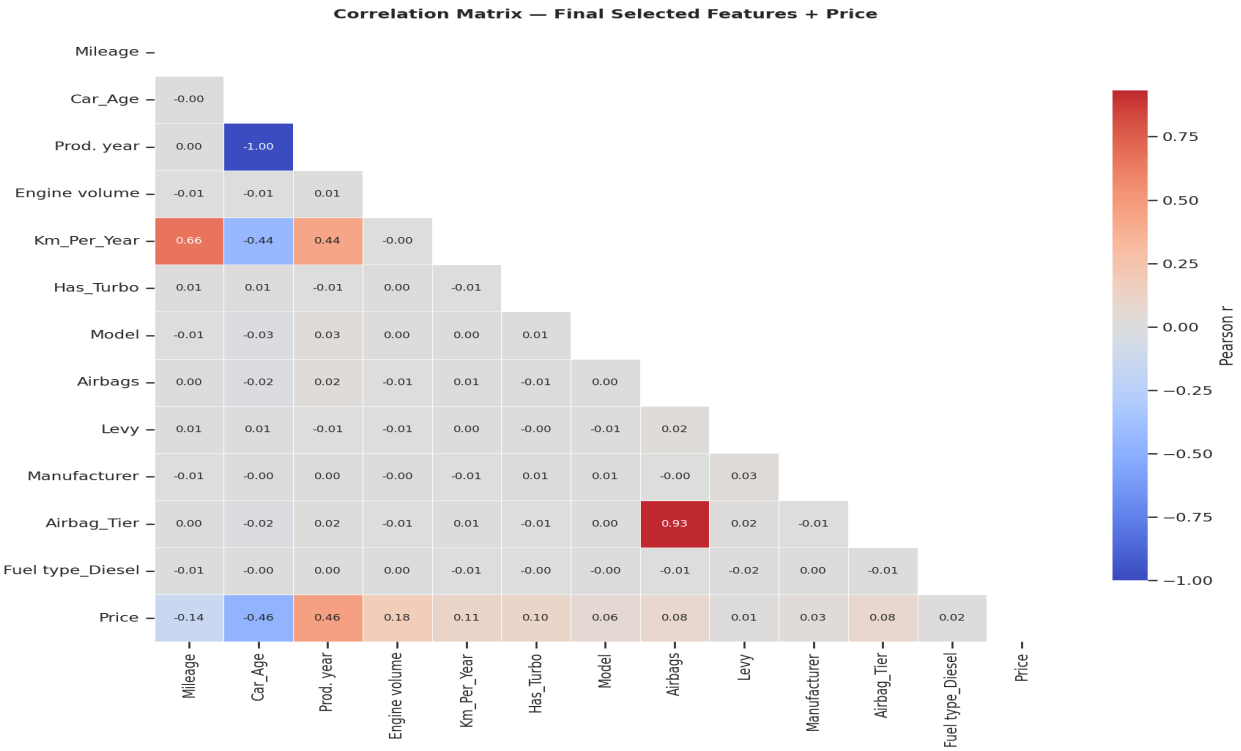
■ Fix 2 — Feature Selection

Problem: Previous threshold (importance > 0) kept all features — no actual dimensionality reduction occurred.

Fix applied: New threshold: importance > 0.002. Result: 14 features kept, 22 dropped.



Feature importance chart with threshold line. Green = selected (14), Grey = dropped (22).



Final correlation matrix of selected features + Price.

CONCLUSION

This milestone successfully transformed the raw car listings data into a clean, machine-learning-ready dataset. Three issues from the initial submission were corrected: target encoding now provides a meaningful numeric signal for high-cardinality features; a principled importance threshold reduces the feature set to the most predictive columns; and a sklearn Pipeline ensures preprocessing is reproducible and leak-free. The processed data is exported and ready for model training in Milestone 3.